DATAVERSE COMMUNITY MEETING

10 Years Sharing Data with Dataverse

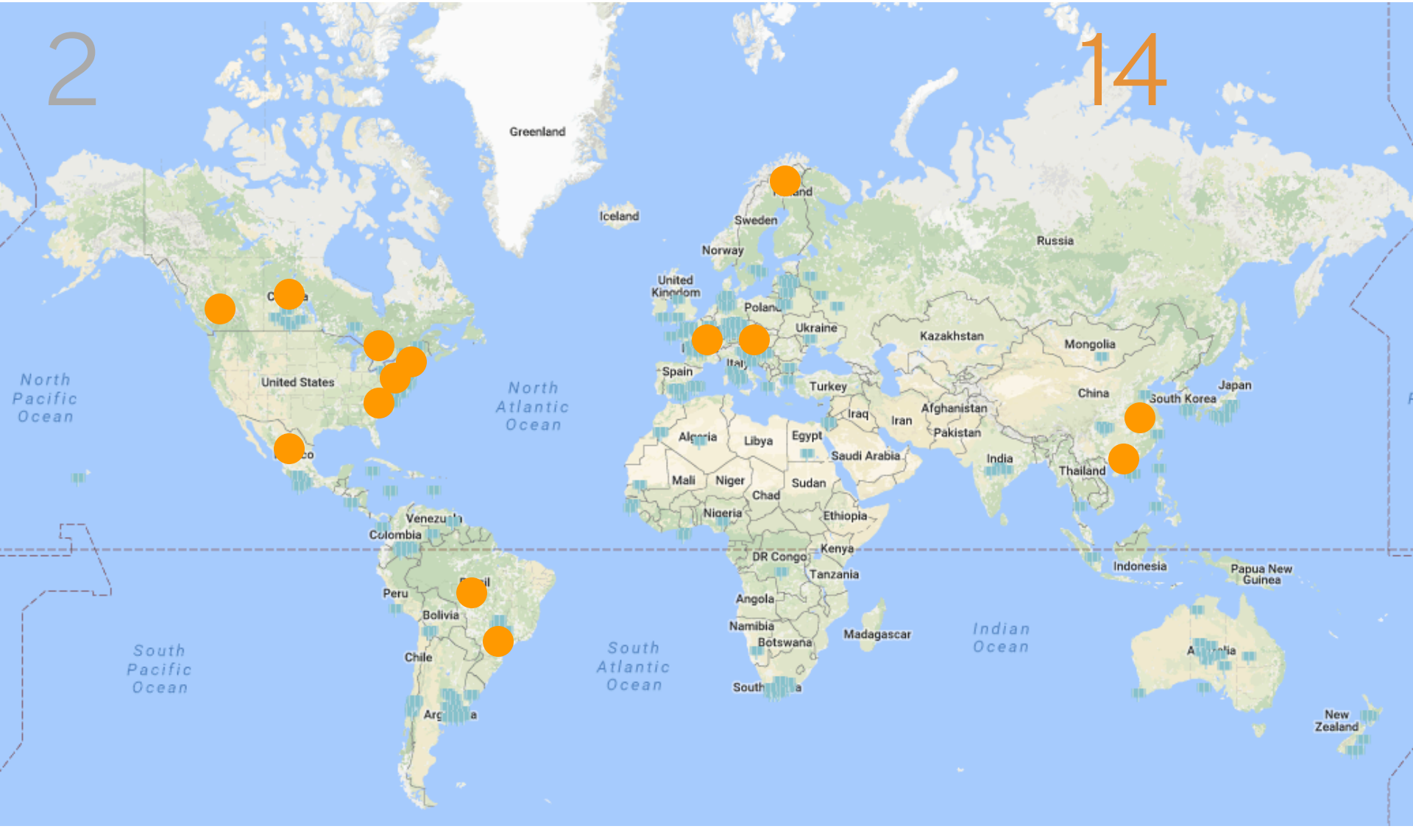
#dataverse2017

<2006

Once there was the VDC

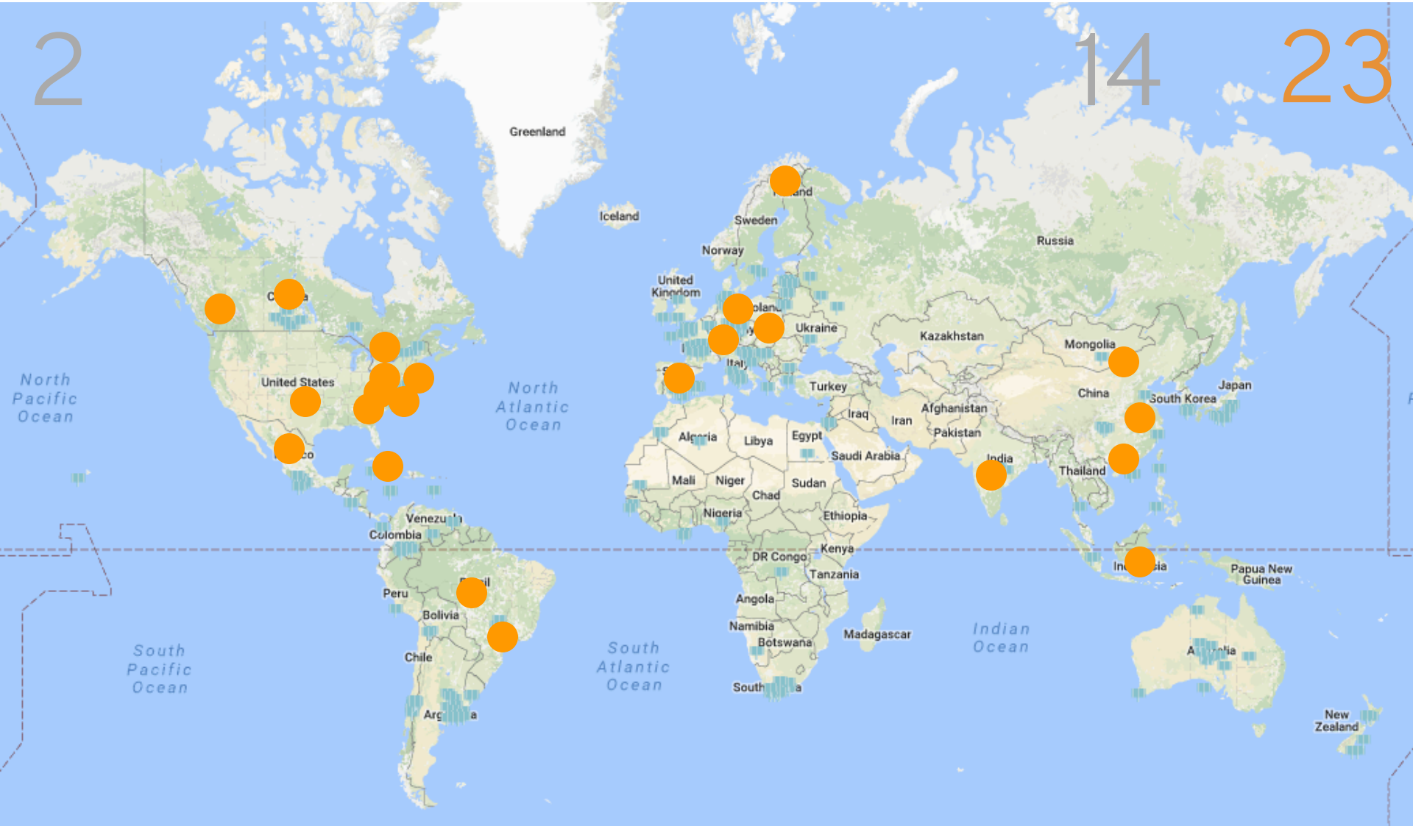
2006 2015

And then came the Dataverse Network



2006 2015 2017

Now we have the Dataverse



RESEARCHERS ARE SHARING AND USING DATA

Harvard Dataverse

> 70,000 datasets

200 datasets/month

> 340,000 files

4,000 files/month

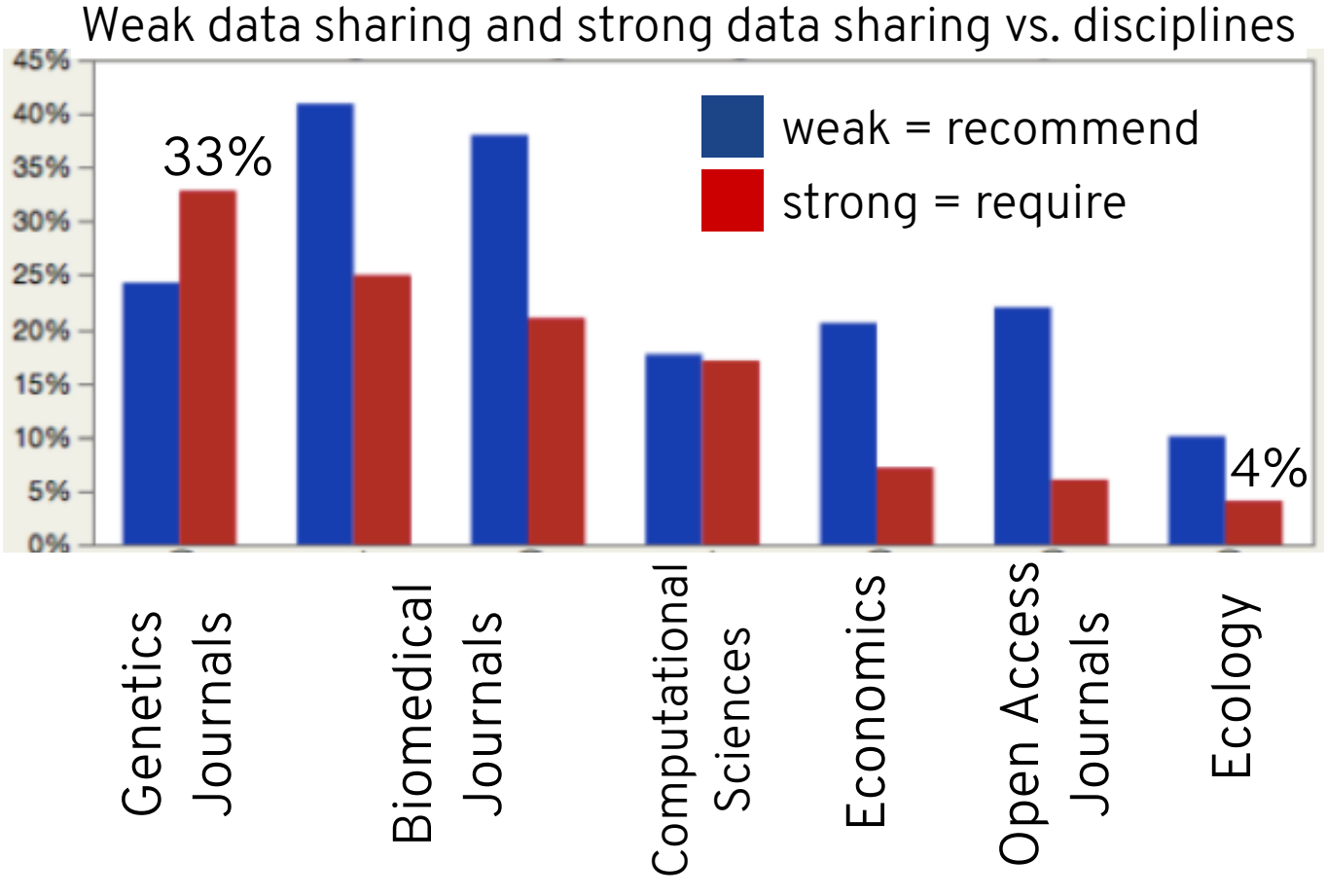
> 2.5 M downloads

60,000 downloads/month

< 2006

When we started, there were
very few journals with data policies,
no data requirements from funders
Greenland
Iceland
Sweden
Russia
Canada
United States
Mexico
Spain
Ukraine
Mongolia
China
Japan
South Korea
Afghanistan
Iraq
Turkey
Mali
Niger
Sudan
Chad
Nigeria
Ethiopia
Tanzania
DR Congo
Angola
Namibia
Botswana
Madagascar
Indonesia
Papua New Guinea
Colombia
Brazil
Peru
Bolivia
Chile
New Zealand
North Pacific Ocean
North Atlantic Ocean
South Pacific Ocean
South Atlantic Ocean
Indian Ocean

2006 2015 2017



Now,
Journals
across disciplines
start supporting data policies

Castro, Crosas, Garnett, Sheridan, Altman, 2017, Journal of Scholarly Publishing, *Forthcoming*

2006 2015 2017

PRIVATE RESEARCH FUNDERS

- Bill and Melinda Gates Foundation Information Sharing Approach
- Sloan Foundation Data Sharing Policy
- Wellcome Trust Data Sharing Policy
- Arnold Foundation
- Moore Foundation
- Robert Wood Johnson Foundation
- HHMI Policy on the Sharing of Publication-Related Materials, Data and Software

PUBLIC RESEARCH FUNDERS

- Department of Agriculture
- Department of Commerce
- Department of Defense
- Department of Education
- Department of Energy
- Department of Health and Human Services
  - Agency for Healthcare Research and Quality (AHRQ)
  - Assistant Secretary for Preparedness and Response (ASPR)
  - Center for Disease Control and Prevention (CDC)
  - Food and Drug Administration (FDA)
  - National Institutes of Health (NIH)
- Department of Homeland Security
- Department of Housing and Urban Development
- Department of Interior
- Department of Labor
- Department of Transportation
- Department of Veterans Affairs
- Environmental Protection Agency (EPA)

And **Funders** require data sharing

WE ARE EXPERIENCING A CULTURAL CHANGE

WE ARE EXPERIENCING A CULTURAL CHANGE

WE ARE THE CULTURAL CHANGE!

2017

King, 1995, Replication, Replication

Altman et al, 2001, A Digital Library for the Dissemination and Replication of Quantitative Social Science

Altman and King, 2007, A Proposed for the Scholarly Citation of Quantitative Data

King, 2007, An Introduction to the Dataverse Network as an Infrastructure for Data Sharing

Crosas, 2012, The Dataverse Network: an open source application for sharing, discovering, and preserving research data

Crosas, 2013, A Data Sharing Story

Altman and Crosas, 2013, The Evolution to Data Citation: from principles to implementation

2014, Joint Declaration of Data Citation Principles

Pepe et al, 2014, How Do Astronomers Share Data?

Goodman et al, 2014, Ten Simple Rules for the Care and Feeding of Scientific Data

Crosas, Honaker, King, Sweeney, 2015, Automating Open Science for Big Data

Castro et al, 2015, Achieving Human and Machine Accessibility of Cited Data

Sweeney, Crosas, Bar-Sinai, 2015, Sharing Sensitive Data with Confidence: The DataTags System

Meyer et al. 2016, Data Publication with the Structural Biology Data Grid Supports Live Analysis

Wilkinson et al, 2016, The FAIR Guiding Principles for Scientific Data Management and Stewardship

Bierer, Crosas, Pierce, 2017, Data Authorship as an Incentive to Data Sharing

The Dataverse project and team leading many aspects of data sharing

AN ACTIVE TEAM AND COMMUNITY

METRICS FROM LAST YEAR, JUNE 2016 TO JUNE 2017

Community

22 COMMUNITY CALLS

190 ATTENDEES
25 ORGANIZATIONS/UNIVERSITIES
10 COUNTRIES

Community

975 GOOGLE GROUP MESSAGES

Community

7,114 IRC MESSAGES

245 UNIQUE USERS

IQSS Dataverse Team

12 SPRINTS

(STARTED IN JANUARY 2017)

IQSS Dataverse Team

220 STANDUP MEETINGS

IQSS Dataverse Team

52,000 SLACK MESSAGES

Code

43 GITHUB CONTRIBUTORS

Code

334 PULL REQUESTS

Code

8,335 GITHUB COMMITS

Support

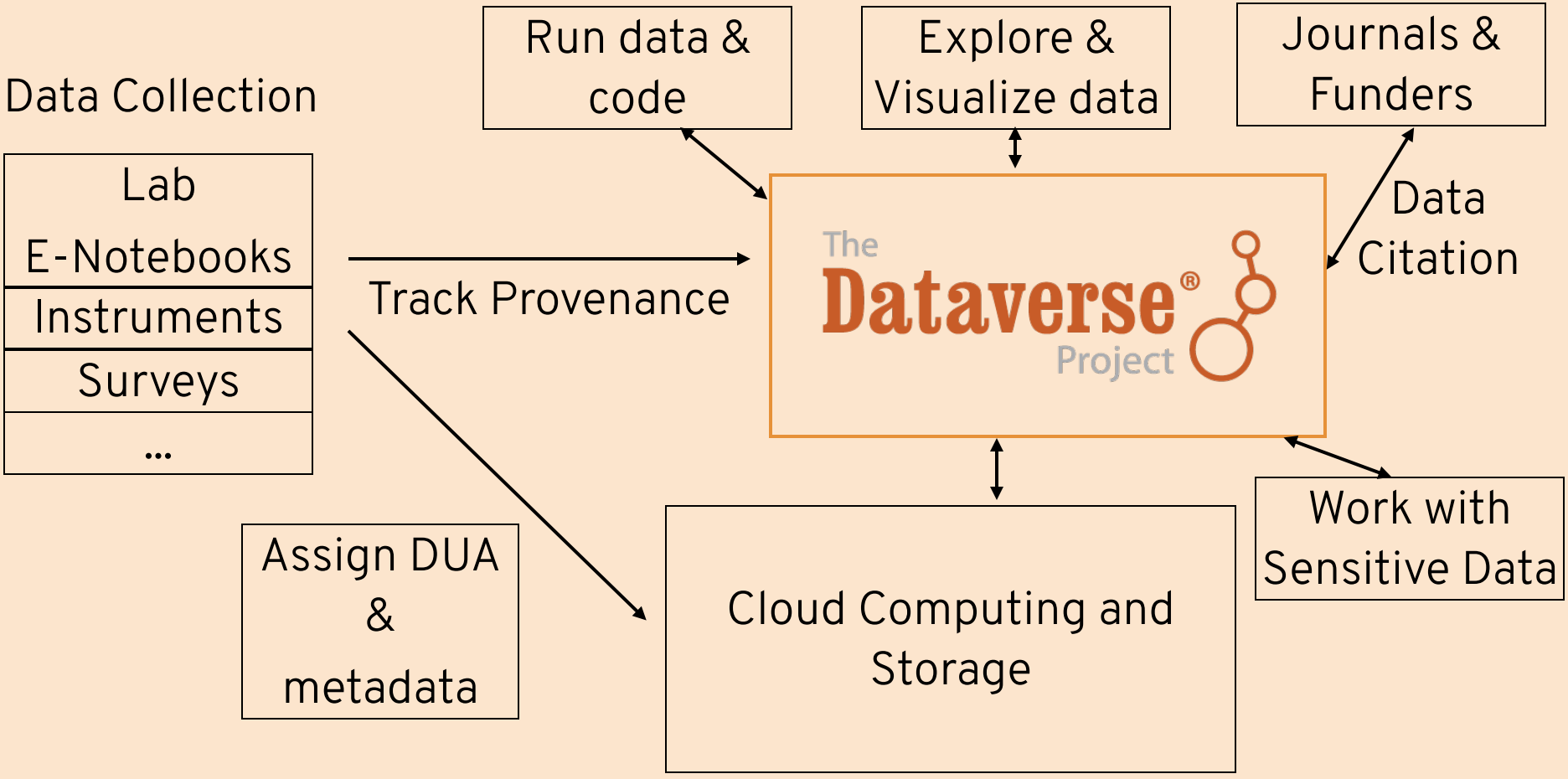
1,153 SUPPORT TICKETS

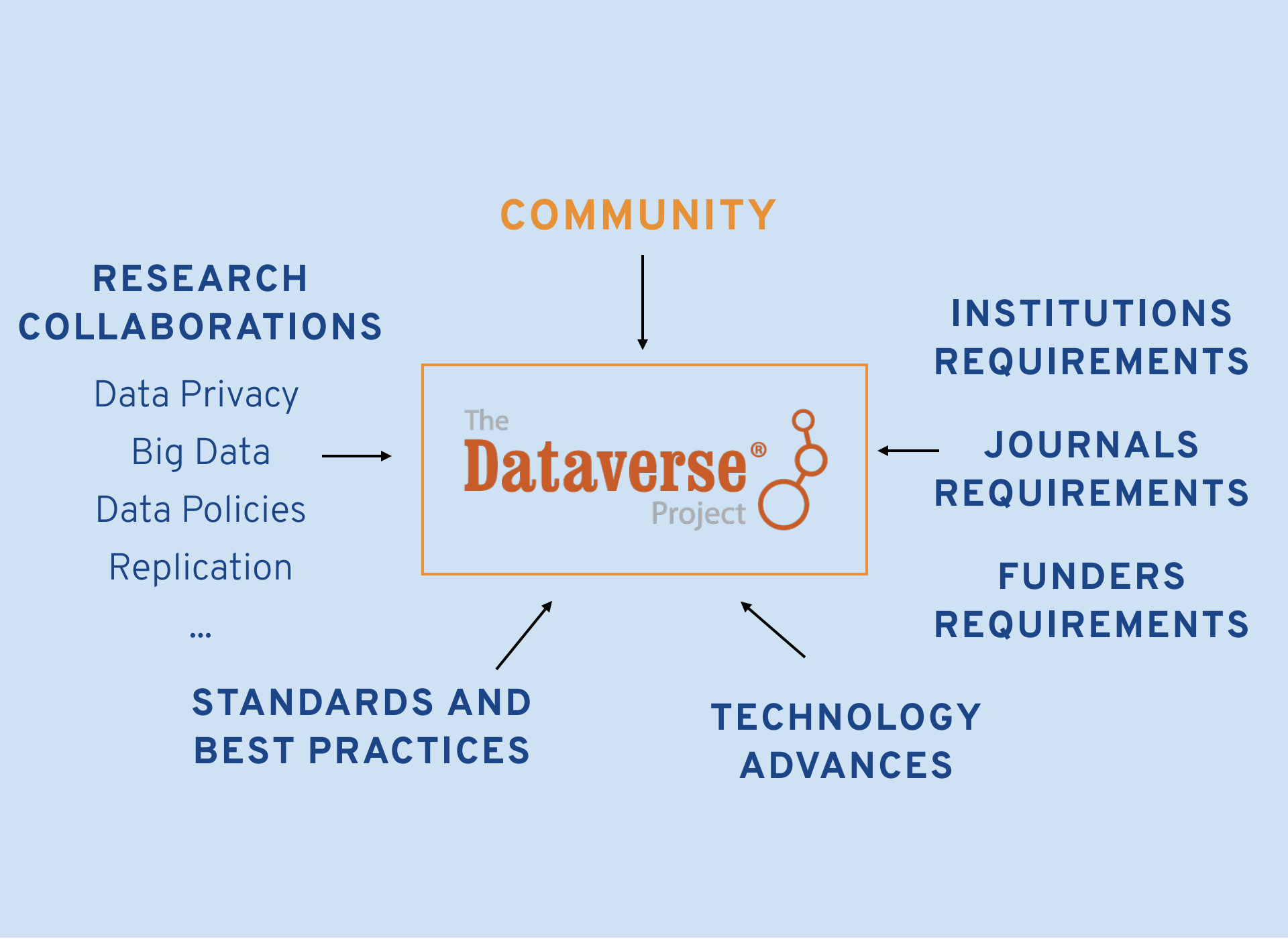
DATAVERSE CUP 2017

A VISION: DATAVERSE AS A KEY PART OF THE FULL RESEARCH DATA LIFECYCLE

TOWARDS A DATA-CENTRIC RESEARCH LIFECYCLE

FROM DATA COLLECTION, TO COMPUTING AND SHARING




COMMUNITY
RESEARCH COLLABORATIONS
Data Privacy
Big Data
Data Policies
Replication
...
The Dataverse® Project
INSTITUTIONS REQUIREMENTS
JOURNALS REQUIREMENTS
FUNDERS REQUIREMENTS
STANDARDS AND BEST PRACTICES
TECHNOLOGY ADVANCES